Fiscal Year:	FY 2026	**Task Last Updated:**	FY 10/02/2025
PI Name:	Pikovski, Igor Ph.D.		
Project Title:	Atomic Quantum Networks on Curved Space-Time		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	07030	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 Physical Sciences NNH22ZDA001N-FP E.6 Fundamental Physics
Start Date:	01/13/2025	**End Date:**	01/12/2028
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Per the Principal Investigator, the period of performance has changed to 01/13/2025 - 01/12/2028. The original period of performance was listed as 12/01/2024 - 11/30/2027 (Ed., 10/3/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Performance Goal No.:			
Performance Goal Text:			

	Einstein’s theory of general relativity is well-tested: ground-based and space-based experiments have confirmed many aspects of the theory, such as the equivalence principle, the redshift, or gravitational waves. Nevertheless, the interface between general relativity and quantum mechanics remains completely unexplored and is one of the main challenges of modern physics. In this proposal we will theoretically study quantum dynamics on curved space-time, focusing on networks of cold atoms and how to use them in space for first experimental tests of this interface between the two theories. The project builds on recent major advancements in both theory and experiment: Experimentally, immense progress has been made to control and deploy cold atom technology for sensing tasks, especially optical lattice atom clocks. These systems provide extremely accurate time measurements and signal sensitivity, and they also allow for versatile quantum control and manipulation of spatial and internal degrees of freedom. On the theory side, the project will build on recent deeper insights on the influence of time dilation on quantum composite systems. In a series of pioneering papers, co-authored by the Principal Investigator (PI) of this proposal, it was shown that new quantum effects arise when composite quantum systems are coupled to post-Newtonian gravity: time dilation causes entanglement between internal clock states and external path superpositions, which can cause a modulation or loss of spatial coherence. This effect is only observable if both quantum theory and gravity beyond the Newtonian limit are simultaneously relevant. These theory results showed a new and promising way to test the gravity-quantum interface. Nevertheless, despite growing interest and variations of the proposed scheme for different matter-wave interferometers, reaching the relevant experimental scales remains a challenge, as large superpositions must be maintained.
Task Description:	Synergizing these theoretical and experimental advancements, this proposal will study and design new tests of quantum dynamics on curved space-time: with networks of entangled atomic clocks in space. The central goal of the proposal is to show how quantum protocols with entangled cold atoms in space can test superpositions of proper time – a unique joint consequence of quantum and gravitational physics. Instead of focusing on matter-waves as in all previous studies, in this project we will explore how entangled networks of atoms can test the same and other new aspects of this interface. The main advantages of networks of clocks are that these highly entangled systems can be safely distributed across large distances, retaining their internal coherence across regions that can significantly differ in space-time curvature. However, a major challenge is to isolate how the genuine interface between gravity and quantum physics can be probed. For entangled clocks, the positions are still classically well described; thus, protocols must be designed to engineer observables where both quantum and general relativistic effects can be tested. We will use quantum control to hybridize different degrees of freedom and systems to propose such measurements and how they can be realized. We will focus on entangled atomic clocks that are also locally controlled for on-demand photon emission and absorption, thus hybridizing atomic and photonic degrees of freedom. These will be used to design protocols in which general relativistic effects are acquired by the entangled clocks, but then coherently transferred onto other systems for which quantum interferometry can be performed. We will analyze what aspects of quantum physics on curved space-time can be probed in such systems and pave the way for future space missions for first demonstrations of this entirely untested domain of physics. Overall, the theoretical results will provide transformative new opportunities to test fundamental physics, exactly in line with NASA interests in fundamental research.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	• The outcomes will enable first tests of quantum theory of curved space-time. • Our work will outline concrete implementations with current and future quantum technologies, for near future space missions and proof-of-principle ground demonstrations. • Will provide new use-cases for quantum networks for fundamental research, enlarging the utility of novel quantum technologies.
Task Progress:	Some of the most counter-intuitive predictions of quantum mechanics, such as the notion of entanglement, have been experimentally tested at increasingly larger distances and system sizes, building confidence in the theory. Similarly, exotic predictions of the theory of general relativity, such as gravitational waves or the gravitational redshift, have been confirmed to great precision. There is, however, a region of the theory which remains fully untested: the intersection of quantum mechanics and general relativity. The main reason for this is the technical difficulty of keeping quantum mechanical effects functioning over the large distances at which gravity has a notable effect. In our research, we have leveraged the rapid development of quantum networking technology to propose near-term tests of the interplay between general relativity and quantum mechanics. The ability to entangle quantum memories over large distances has been demonstrated in numerous experiments in recent years. This is driven by envisioned applications of large-scale quantum networks such as secure communication and distributed quantum computing – a “quantum internet”. We show that a quantum network also brings new opportunities for fundamental tests of nature. We demonstrate that entangled quantum memories can function as de-localized quantum clocks, which can probe and interfere different regions of space-time with varying time-flows, amounting to tests of the interplay between quantum mechanics and general relativity. Gravity affects the entangled clocks by changing the flow of time, while quantum mechanics predicts such a varying flow of time to exist in superposition. Our protocols show that this interplay of the two effects can be probed with near-term experiments on Earth and in space. Our work combines techniques from quantum networking, quantum enhanced metrology, and distributed quantum computing to offer a first avenue to test quantum theory on curved spacetime. It opens a new avenue for fundamental tests with quantum networking technology, and may reveal whether our current theories survive or whether unexpected new physics emerges.
Bibliography Type:	Description: (Last Updated: 10/08/2025)
Articles in Peer-reviewed Journals	Borregaard J, Pikovski I. "Testing quantum theory on curved space-time with quantum networks." Phys. Rev. Research. 2025 May;7(2):023192. https://doi.org/10.1103/PhysRevResearch.7.023192 , May-2025
Articles in Peer-reviewed Journals	Covey JP, Pikovski I, Borregaard J. "Probing curved spacetime with a distributed atomic processor clock." PRX Quantum. 2025 Jul 1;6(3):030310. https://doi.org/10.1103/q188-b1cr , Jul-2025

Significant Media Coverage	Minic D. (feature of Covey JP, Pikovski I, and Borregaard J research) "Testing quantum theory in curved spacetime. Featured in the publication of the American Physical Society." Physics. 2025 Jul 21;18:135. https://physics.aps.org/articles/v18/135 , Jul-2025
Significant Media Coverage	Hossenfelder S. (on paper written by JP Covey, I. Pikovski and J. Borregaard) "This clever experiment could finally advance physics. " YouTube Science channel: https://youtu.be/7I4Nuu-t2JU?si=tkVxfRX0izDQ4PN7 , Sep-2025